

***REJUVENATION STRATEGY OF DYING
SPRINGS AND NAULAS IN DHARGAD RAINFED
RIVER WATERSHED, JAIGAN VALLEY,
DISTRICT BAGESHWAR, KUMAUN HIMALAYA,
(UTTARAKHAND)***

***PHASE-I
REJUVENATION OF SPRINGS AND NAULAS
OF JATHA AND PASS VILLAGES
IN DHARGAD WATERSHED***

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Executive summary

The Himalayan region, known as the "Water Tower of Asia," is experiencing a severe crisis both in glacial-fed and non-glacial rain fed river watersheds. Rain fed rivers in Uttarakhand play a vital role in supporting agriculture, industry, and human consumption, having over 353 significant rain fed rivers critical for local communities. Within the watersheds of these rainfed rivers, about 70% villages and more than 80% urban centers of the Uttarakhand state are located. Due to anthropogenically and technogenically accelerated climate change impacts, the water sources such as springs, streams are steadily transforming from their perennial nature to ephemeral nature, and the major rivers are dwindling very fast due to decrease in base flow during summer season. Such perturbed hydrological character in the region is very critical for local communities as the environment, rural livelihoods, agriculture, and ecosystems are being adversely affecting. Failing to take an planned scientific action for groundwater augmentation through massive rainwater harvesting mechanical and biological measures in the recharge areas of springs and streams -will lead to dire consequences, including severe water scarcity, biodiversity loss, and exacerbated climate change impacts.

How to rejuvenate the dying springs/stream and steadily dwindling rivers of the rain fed watersheds ? This report is a modest attempt towards addressing this crucial problem which aims to develop a model of scientific integrated strategy for rejuvenation of springs, streams and rivers. For this purpose, a rain fed river watershed of the Kumaun Lesser Himalaya (in district Bageshwar, Uttarakhand State), viz., the Dhargad river watershed has been employed as a natural laboratory. The fundamental objectives of this work are as follows:

- i) Identify and prioritize dead and dying springs, and streams in the watershed.*
- ii) Assess the status of springs and streams and identify potential recharge areas. Develop a*
- iii) GIScience based decision support system for rejuvenation planning. Implement and*
- iv) monitor rejuvenation measures in association with local inhabitants and Gram Panchayats to improve water availability and reduce land degradation and improve the economy of the local inhabitants. To mitigate the adverse impacts of climate change in hydrology, in particular, and the physical and human environment in general.*
- v)*

AGIS-based strategy aims to identify and rejuvenate dead and dying water sources, with objectives including the assessment of springs, prioritizing recharge areas, and community engagement in implementation. The project employs GIS technologies, remote sensing, and community involvement to develop a rejuvenation plan that includes mapping water sources and planning interventions. The rejuvenation plan provides micro-plans both mechanical treatments (like infiltration holes, infiltration trenches, bio-plugs and check dams etc) and biological treatments (such as afforestation and nursery development etc) to restore the watershed. Effective implementation of the rejuvenation measures requires collaborative efforts with local communities and government agencies to ensure sustainable practices. This plan emphasizes a phased approach to address the urgent needs of the local water resources, beginning with the experimental areas of Jatha and Pass villages. Implementing these measures will enhance groundwater availability, reduce land degradation, mitigate the process of adverse impacts of climate change and improve economic conditions for local inhabitants while promoting sustainable water management practices.

1.0 INTRODUCTION

1.1 The Crucial Problem The Himalayan region, often referred to as the "Water Tower of Asia," is home to numerous rivers, springs, and naulas that support millions of people downstream. However, both the glacial fed and the rainfed river regimes of the Himalaya are facing an unprecedented crisis¹⁻⁴. Springs, naulas and streams are drying up at an alarming rate, threatening the very fabric of rural livelihoods, agriculture, and ecosystems. The spatial distribution of rainfed rivers reveals that there are as many as 353 rain fed streams/rivers of more than fourth order, having a total length of about 5796km⁵. More than 45% of the geographical area of Uttarakhand is drained by non-glacial rain fed rivers wherein about more than 70% of villages more than 80% of urban centers are located. This rainfed rivers play a crucial role in supporting agriculture, industry, and human consumption in Uttarakhand. The major rainfed river systems of Uttarakhand are Kosi, Gaula, Saryu, western Ramganga, western and eastern Nayar etc.

The Dhargad River watershed (Fig.1), a tributary watershed of the Jaigan River which finally joins the Saryu River at Lweta village (elevation 716m, 29°42'13.551"N Lat and 79°53'40.896"E Long) in the Bageshwar district, is a classic example of rain fed river which is steadily dwindling and dying. This rain fed watershed is witnessing a steady decline in groundwater storage, leading to the transformation of naulas, springs and streams from their perennial to intermittent and ephemeral nature.

1.2 The Consequences of Inaction are Dire

If no groundwater augmentation measures for rainwater harvesting through the mechanical treatment measures were taken in this rain fed watershed, the consequences of this inaction are dire which will result in severe water scarcity, ecosystem degradation and adverse impacts of climate change impacts.

1.2.1 Water Scarcity

Declining water sources will exacerbate drinking water scarcity, impacting human consumption, agriculture, and livestock.

1.2.2 Ecosystem Degradation

Dying springs and streams will lead to loss of biodiversity, ecosystem disruption, and reduced agricultural productivity.

1.2.3 Climate Change

The Himalayan region is highly vulnerable to climate change⁶⁻⁹, which will further exacerbate water stress and ecosystem degradation.

The need of the hour is to address the problem of dying springs, naulas, and streams in the Dhargad River watershed. This GIS-based rejuvenation strategy helps in identifying priority areas for intervention, optimize resource allocation, and ensure sustainable management of these vital water sources.

2.0 OBJECTIVES

To address the above mentioned crucial problems, this project has been proposed to make a scientific attempt using GIScience10-13 technologies i.e., Remote Sensing (RS) Geographic Information System (GIS), Global Positioning System (GPS) and Digital Cartography to rejuvenate phase wise, the dying naulas, springs and streams of this rain fed experimental watershed, viz., the Dhargad Watershed (Fig.1), The fundamental objectives of this project are as follows.

1. Identify and prioritize dead and dying naulas, springs, and streams in the watershed.
2. Assess the status of springs and streams and identify potential recharge areas.
3. Develop a GIS-based decision support system for rejuvenation planning.
4. Implement and monitor rejuvenation measures in association with local inhabitants and gram panchayats to improve water availability and reduce land degradation and improve the economy of the local inhabitants.

3.0 ABOUT THE DHARGAD RIVER WATERSHED

The study area, viz., the Dhargad watershed lies in between 29°43'38.117" N to 29°44'53.908" N Lat and 79°43'39.756" E to 79°44'47.547" E Long in the Lesser Himalayan rainfed river regime in district Bageshwar (Fig.1), which encompasses an area of 2.32 km². The Dhargad is a third order influent stream which drains her water to the Ganga River through the Jaigan, Saryu and the Kali River. The Dhargad stream joins the Jaigan river near 29°44'48.51" N Latitude to 79°44'41.773" E Longitude at an elevation of 1225m which meets with the Saryu River near 29°42'16.183" N Latitude to 79°53'41.713" E Longitude at an elevation of 726m from msl. The altitude in the watershed varies in between 1295 m to 1492m. Climatically, the watershed falls in cool temperate and cold temperate climatic regions¹⁴. Geologically, the watershed is composed of metamorphosed and sedimentary rocks of the Damta group¹⁵⁻¹⁶ _

The users of water resources of this watershed are of two villages. These are village Jatha and Pass. For rejuvenation of dead and drying springs and Naulas, these two villages have been identified as experimental areas for the first phase work in this watershed (Fig.2).

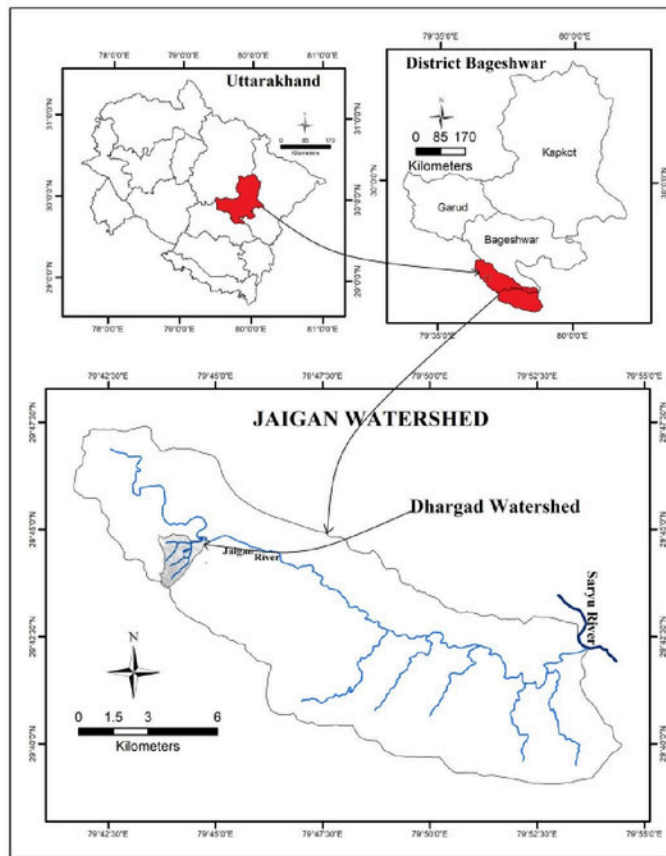


Fig-1: Location of the Dhargad watershed in the Jaigan River Watershed, District Bageshwar, Kumaon Himalaya.

4.0 METHODOLOGY

For preparing a scientific strategy for rejuvenation of dead and dying water sources of the phase-I experimental area in village Jatha and Pass, the technologies of GI Science, i.e., Remote Sensing, GIS, GPS and Computer Cartography have been used which was further followed by field surveys with the personnel of AMAN NGO and the local Gram Panchayats. For deliberations and discussions on the naulas, springs and streams rejuvenation strategies, meetings were organized with the members of villages of the experimental areas, viz., Jatha and Pass. The work plan of rejuvenation of naulas, springs and streams were finally worked out in this meeting of village panchayat. A brief account of GIS and Remote Sensing based scientific methodology adopted in this work is divisible into four groups as presented in the following paragraphs.

4.1 GIS based Mapping of the Phase-I Experimental Recharge Area

- i) Data Collection: Gather satellite imagery, topographical maps, and hydro-meteorological data for the experimental recharge area.
- ii) GIS Analysis: Delineate the watershed area, identify recharge zones, and analyze land use/land cover, slope, and drainage patterns.
- iii) Field Surveys: Conduct field visits to validate GIS analysis and identify potential rainwater harvesting sites.
- iv) Rejuvenation Plan: Develop a comprehensive plan incorporating rainwater harvesting structures, groundwater recharge measures, and watershed management practices.

v) Deliberation and Discussions with Village Panchayat (ANNEXURE -I): The developed rejuvenation strategy was presented among the members of Village Panchayat Jatha and Pass and after deliberation and discussion, the rejuvenation action plan was worked out with concurrence of the Village Panchayat¹⁷.

4.2 GIS-based Rejuvenation Plan

- i) Recharge Zones: Identify and prioritize areas for rainwater harvesting and groundwater recharge.
- ii) Rainwater Harvesting Structures: Propose construction of infiltration holes, infiltration trenches, bio-plugs and check dams in the experimental recharge area.
- iii) Watershed Management: Implement soil conservation measures, afforestation, plantation and sustainable land use practices.
- iv) Groundwater Augmentation: Design artificial recharge systems to enhance groundwater levels.
- v) Monitoring Network: Establish a monitoring system to track water levels, flow, and quality.

4.3 Implementation Strategy

- i) Community Engagement¹⁷: Involve local communities in planning, implementation, and maintenance of rejuvenation measures.
- ii) Government Collaboration: Collaborate with government agencies for technical support, funding, and policy backing.
- iii) Phased Implementation: Implement the plan in phases, prioritizing high-impact area

PHASE –I

REJUVENATION OF SPRINGS AND NAULAS

OF EXPERIMENTAL RECHARGE AREA

IN JATHA AND PASS VILLAGES

1.0 INTRODUCTION

Skilled human resources and sufficient funds are needed for reversing steady transformation naulas, springs and streams for their perennial nature to ephemeral nature¹⁸. There is acute scarcity of both. In view of this acute scarcity of skilled human resources and sufficient funds, the work of rejuvenation of the Dhargad watershed resources must be carried out in different phases. In the first phase – as per the need of the local village Panchayat Jatha and Pass villages¹⁷, a strategy has been developed by delineating recharge zone to rejuvenate the dead and dying naulas, springs and streams of these two villages (Jatha and Pass).

2.0 ABOUT THE EXPERIMENTAL AREA PHASE-I

The experimental area where the first phase work must be carried out, lies between 29° 44' 20.846" N to 29° 44' 36.699" N Lat and 79° 44' 7.329" E to 79° 44' 33.371" E Long which encompasses an area of 0.17 km² in the Dhargad River watershed (Fig.2).

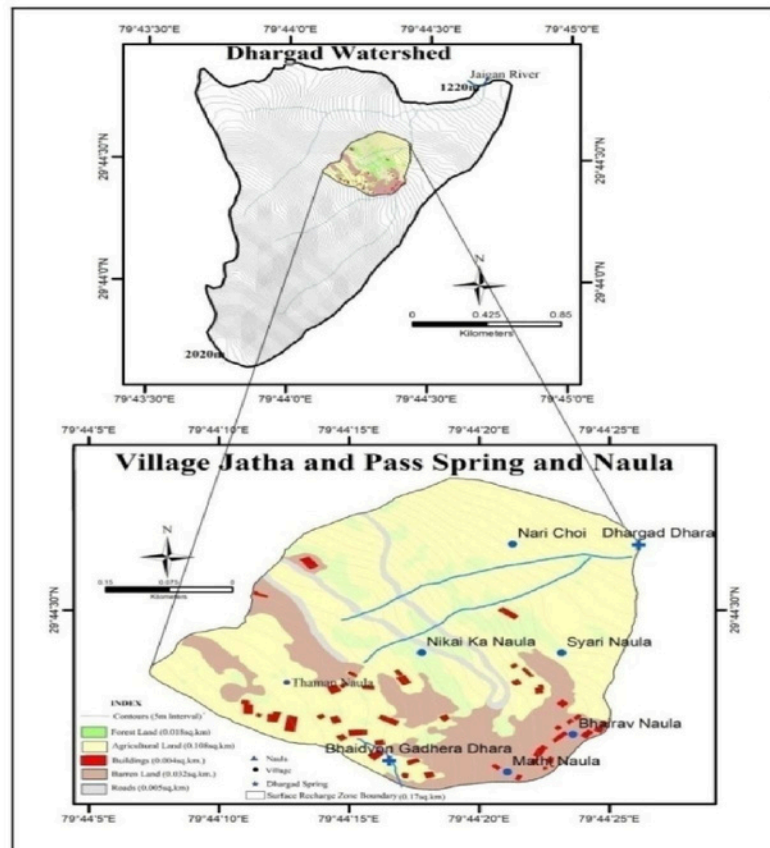


Fig.2: Location of the Phase-I Experimental area for rejuvenation of naulas, spring and stream in Village Jatha and Pass

3.0 PHYSICAL GEOGRAPHY

A brief account of physical geography (of the experimental area of the first phase work) which includes geology, climate and weather, hydrology and geomorphology, is presented in the following paragraphs.

3.1 Geology

Geologically, the rocks of the study area are composed of the Raut Gara Formation¹⁴ of the Damta Group (Fig.3) made-up of dominant quartzites often with feldspar and mica minerals; interbedded with fine grained foliated metamorphic phyllite with silky sheen; and fine-grained, low-grade metamorphic slate rocks¹⁵⁻¹⁶. These rocks have undergone significant tectonic activities, including metamorphism and thrust deformation, therefore are highly jointed and fractured and shared at places.

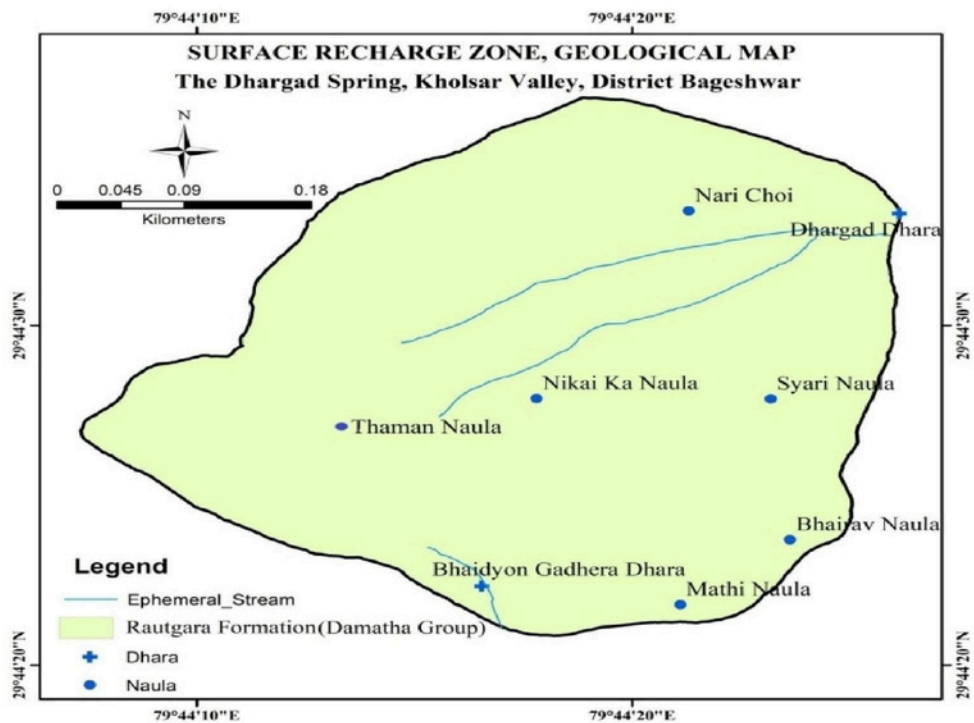


Fig.3: Lithological formation map of the experimental area

3.2 Weather and Climate

Climatically, the experimental area falls under cool temperate climate zone¹⁴ (Fig.4) which is typically found in elevations between 1,300 to 2,000 m from msl. The average annual temperature in the cool temperate climate zone ranges from 10°C to 18°C. Winters are cold, with temperatures often dropping to 0°C or below, while summers are mild, with temperatures up to 25°C. The annual rainfall is around 1,000 to 1,500 mm, with most of it received during the monsoon season (June to September).

The summer season (March to May) is mild, with temperatures ranging from 15°C to 30°C. Winters (October to February) are cold, with temperatures often dropping to 0°C or below. It's an ideal time for snowfall enthusiasts, with December and January being the peak months. Rainy Season (June to September) brings moderate rainfall, with temperatures around 20°C.

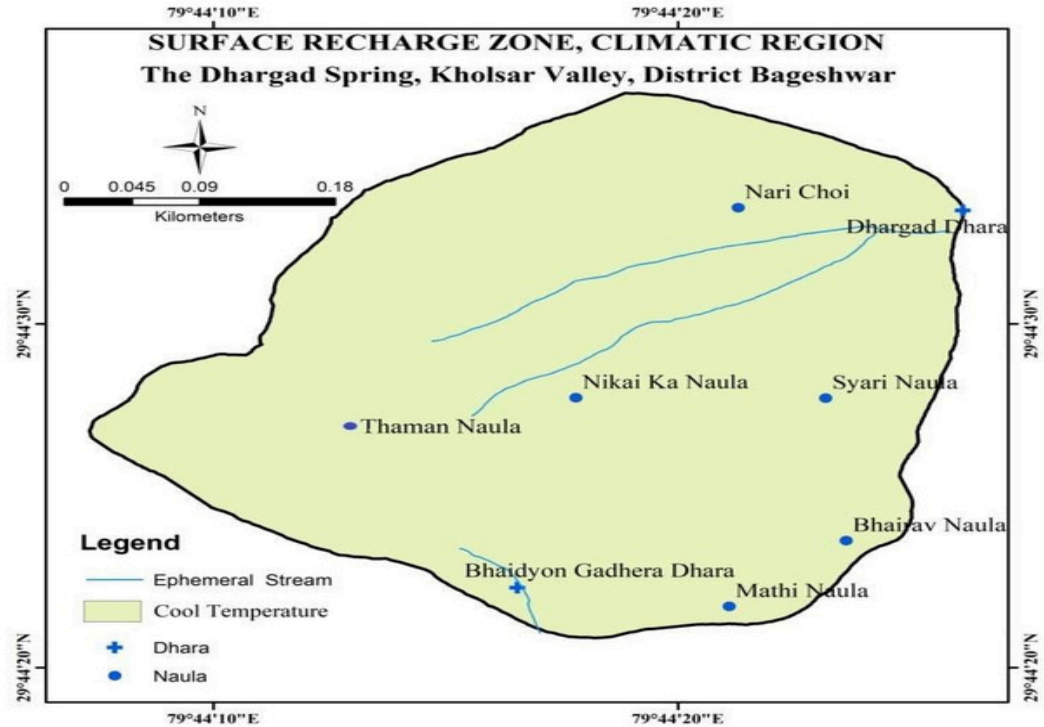


Fig.4: Cool Temperate Climatic region in experimental recharge area.

3.3 Surface Hydrology

In the experimental area, there are as many as seven traditional drinking water sources out of these five are naulas and two are springs. Besides these, there are two ephemeral streams also, which drain their water during monsoon rains in the Dhargad Stream. Figure 5 depicts the spatial distribution of all these surface water sources of the experimental area and a brief account of all these surface water sources is presented in the following paragraphs.

3.3.1 Springs and Naulas

There are seven traditional natural water sources of the study area divided into two types, i.e., springs and naulas. The summary of characteristics of these naulas and springs is presented in Table-1 and a brief description of these naulas, and springs is presented in the following paragraphs.

Table-1: Details of springs and naulas of the study area
(Source –Field Survey by Jagdish Chandra and his Team

S N	Name of Water Source	Type	Beneficiary families	Family Members	Water Use	Condition 10 years back	Present condition
1	Dhargad	Spr ing	80	350	Drinking and Working Animals use	Working	Working
2	Bhaidyon	Spring	62	150	Drinking, Agriculture and Animals use	Working	Working
3	S yar i	Naula	12	25	Drinking and Working Agriculture	Working	Working
4	Mathi	Naula	5	11	Drinking, and Working Animals use	Working	Damage
5	Nikai	Naula	Nil	Nil	Nil	Working	Damage
6	Bhairav	Naula	Nil	Nil	Nil	Damage	Damage
7	Thaman	Naula	30	110	Animals use (Only in Summer)	Working	Damage

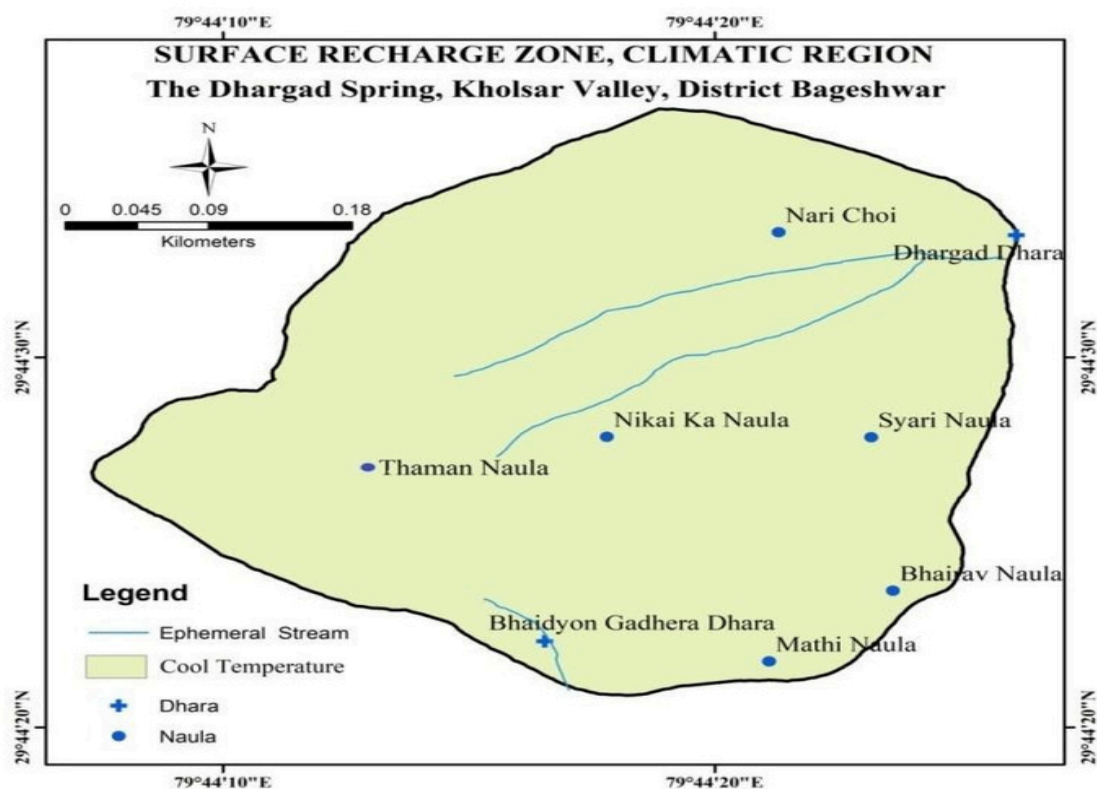


Fig.5: Spatial distribution of springs and naulas in experimental recharge area.

3.3.1.1 Dhargad Spring (Plate-I)

The Dhargad spring (29° 44' 33.419" N Lat and 79° 44' 26.042" E Long) lies at the height of 1324m from the msl in the northern part of the experimental recharge zone area (Fig.5) having discharge capacity of 0.34l/second (in August 2025). Besides drinking, the water of this spring is used for washing clothes, animals and irrigation also. A population of 350 people of 80 families of village Pass and Jatha are users of this spring water for drinking and other uses.



Plate-I: Dhargad Spring of the experimental recharge area.

3.3.1.2 Bhaidyon Spring (Plate-II)

The Bhaidyon spring (29° 44' 22.337" N Lat and 79° 44' 16.526" E Lon) lies at the height of 1441m from msl in the southern part of the experiments recharge zone (Fig.5). Beside drinking, the water of this spring is used for washing clothes, animals and irrigation also. A population of 150 people of 60 families of village Jatha are users of this spring.



Plate-II: Bhaidyon Spring of phase –I experimental recharge area

3.3.1.3 Syari Naula (Plate-III)

The Syari Naula (29°44'27.844" N Lat and 79°44'23.175" E Long) lies at the height of 1362m from msl in the eastern part of the experiments recharge zone (Fig.5). Besides drinking, the water of this spring is used for washing clothes, animals and irrigation also. A population of 25 people of 12 families of village Jatha are user of this naulas.



Plate-III: Syari Naula of the phase –I, experimental recharge area

3.3.1.4 Mathi Naula (Plate-IV)

The Mathi Spring (29°44'21.801" N Lat and 79°44'21.086" E Long) lies at the height of 1397m from msl in the southern part of the experiments recharge zone (Fig.5). This naula is completely damaged. Besides drinking, the water of this spring is used for washing clothes, and irrigation also. A population of 11 people of 5 families of village Jatha are users of this Naula.

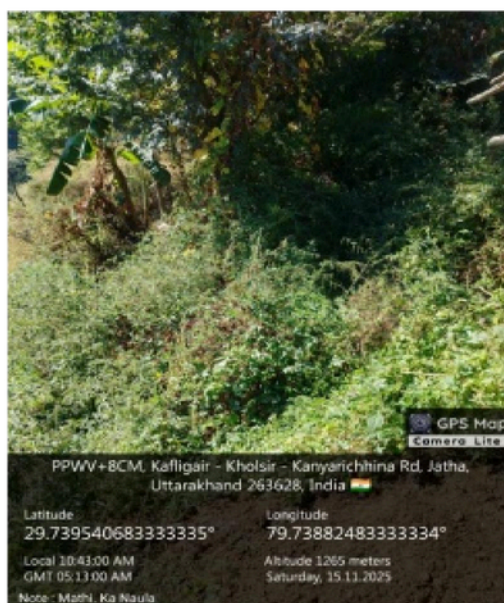


Plate-IV: Mathi Naula of the phase –I, experimental recharge area

3.3.1.5 Nikai Naula (Plate-V)

The Nikai Naula (29°44'27.862" N Lat and 79°44'17.775" E Long) lies at the height of 1306m from the msl in the central part of the experiments recharge zone (Fig.5). These naulas are completely damaged.



Plate-V: Nikai Naula of the phase –I experimental recharge area

3.3.1.6 Bhairav Naula (Plate-VI)

The Bhairav Naula (29°44'23.703" N Lat and 79°44'23.621" E Long) lies at the height of 1397m from the msl in the southeastern part of the experiments recharge zone (Fig.5). These naulas are used for drinking water for animals of 30 families having a population of 110. These naulas are completely damaged.

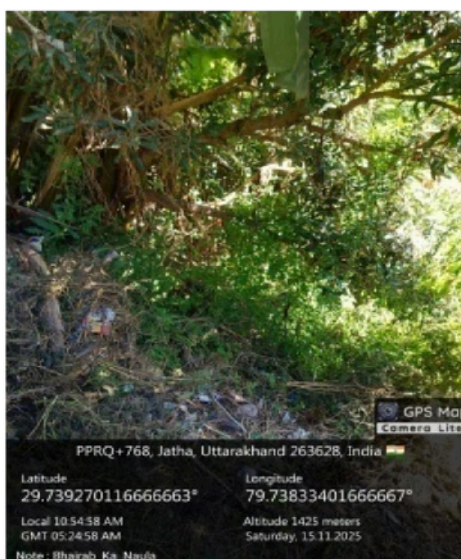


Plate-VI: Bhairav Naula the phase –I experimental recharge area

3.3.1.7 Thaman Naula(Plate-VII)

The Thaman Naula (29°44'27.039" N Lat 79°44'12.802" E Long) lies at the height of 1449m from the msl in the southern part of the experiments recharge zone (Fig.5).

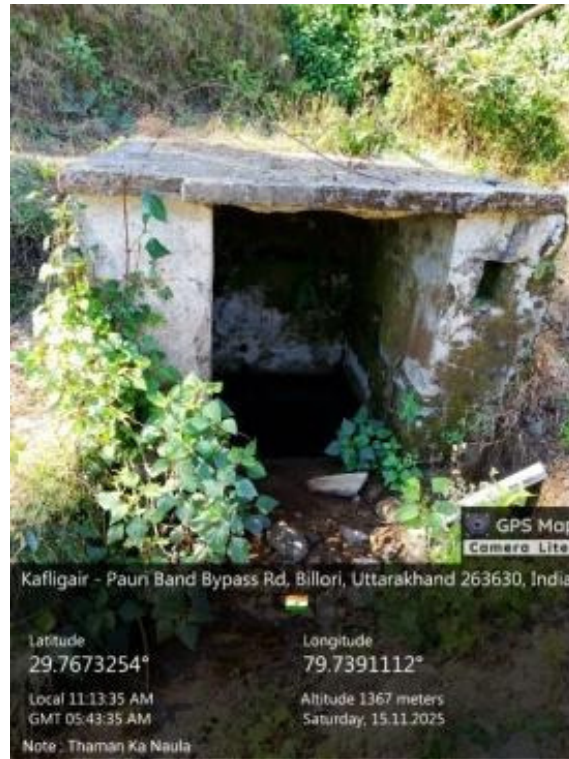


Plate-VII: Thaman Naula the phase –I experimental recharge area

3.3.2 Ephemeral Rills

There are three ephemeral rills in the concave hollows in the experimental area (Fig. 5) which become very active during monsoon rains. These are major sources of generating floods coupled with very high sediment load.

3.4 Geomorphology

The geomorphic study of the experimental recharge area was carried out with the help of the high-resolution contour map (Fig.6) followed by field surveys. Based on topographic map and field survey, a brief account of relief, morphology and hydrography is presented in the following paragraphs.

3.4.1 Relief

The contour map (Fig.6) reveals that the average height of the experimental recharge zone stands at 1850 m which varies in between 1325 m to 1475m. The relief zone map of the recharge zone (Fig.7) reveals that about 17.65 % (0.03km²) area of the recharge zone lies on high relief zone where the height is more than 1400m, about 52.94 % (0.09km²) lies on the medium relief zone where the height varies in between 1350 m to 1400 m and the remaining 29.41 % (0.05km²) area falls under low relief zone where the height is below 1350m (Table-2).

Table-2: Distribution of area under different relief zones in the experimental recharge zone.

Relief Zones in Meters	Area in km ²	Area in %	Remark
<1350	0.03	17.65	Low
1350-1400	0.05	29.41	Medium
>1400	0.09	52.94	High
Total	0.17	100	

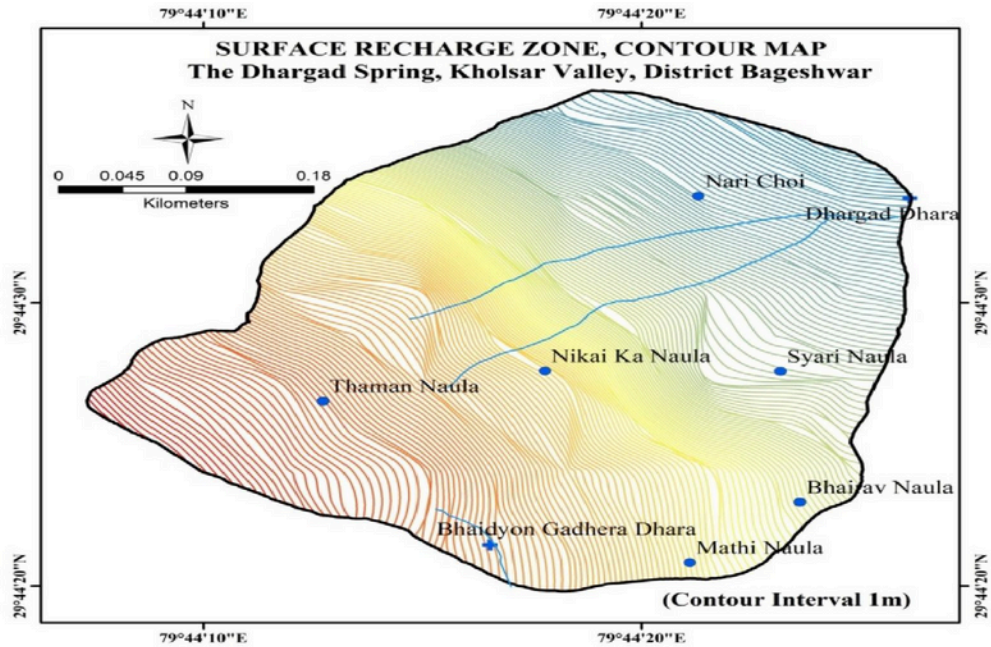


Fig.6: Contour map of the experimental recharge area.

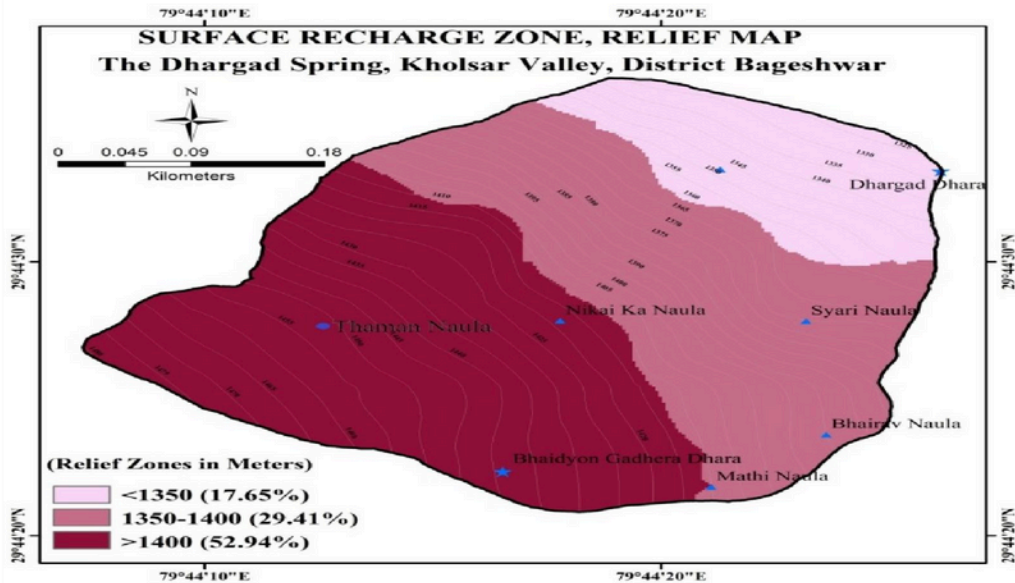


Fig.7: Relief zone map of the experimental area showing different relief zones.

3.4.2 Morphology

For delineating micro level geomorphic landforms, the satellite image of the study area was overlain by the high resolution contours (Fig.8). Using this map (Fig.8), micro-geomorphic features were marked as depicted by figure 10. This morphological map, reveals that the landscape of the study area is mainly carved by the pluvial (rainwater) process of sheet flow (overland flow)) and sheet wash erosion. The pluvial landforms in the study area are (Fig.9):

- i) convex hill slopes(CV),
- ii) concave hill slopes(CC),
- iii) hollows (unchanneled valleys),
- iv) ephemeral rills. and
- v) structural benches of level land

The convex slopes are the recharge zones while the concave slope zones are discharge zones. The convex slopes areas are the most suitable sites for mechanical treatment measures such as infiltration holes and infiltration trenches for rainwater harvesting for groundwater augmentation. This slope form is also suitable for contour trenching and terrace ponds. Wherever agricultural land is available (see Plate-XII and XII). Wherever the agricultural and barren land is available on concave slope, contour trenching and terrace ponds may be constructed for rainwater harvesting.

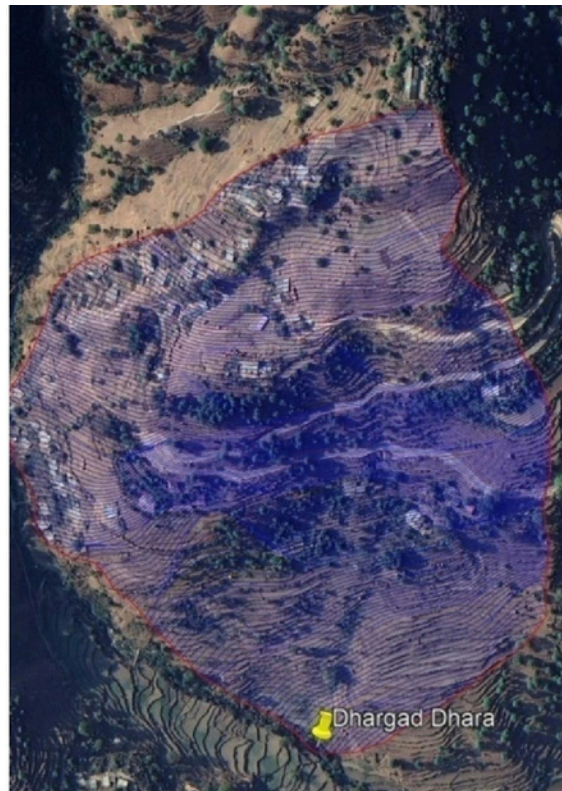


Fig.8: The satellite image of the experimental recharge area overlain by high-resolution contour (1m resolution) map

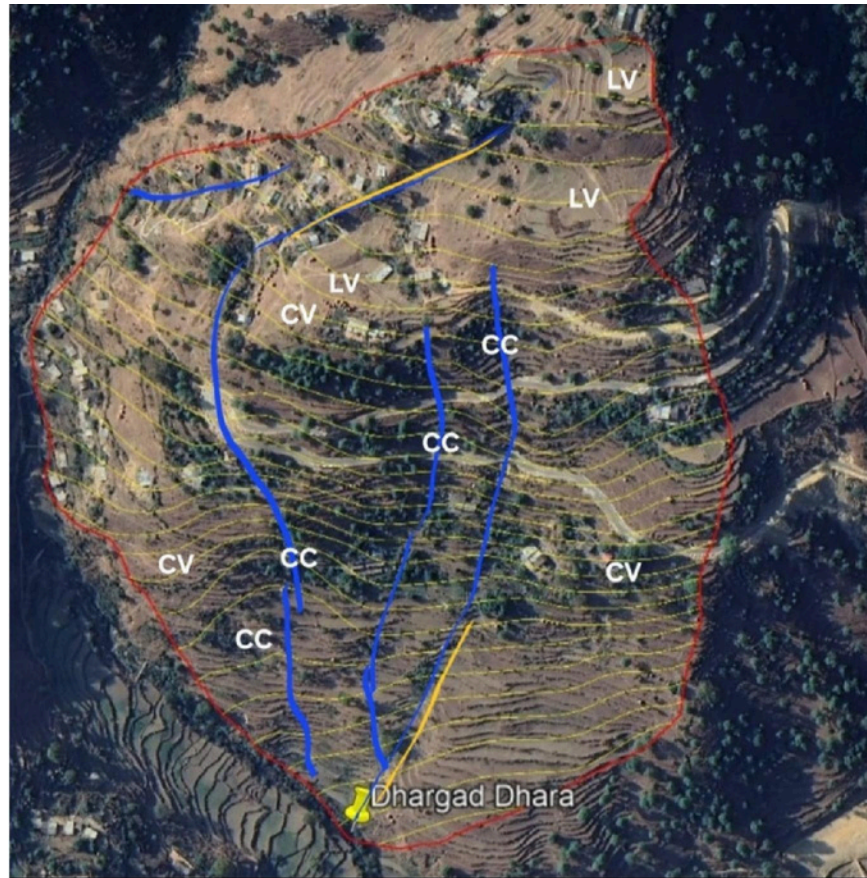


Fig.9:Geomorphological features in the experimental recharge area depicting the locations of different geomorphic features: CV-Convex Slope,CC-Concave slope,LV-Level Slope,Axis of Hollows,ephemeral rills.

4.0 HUMAN GEOGRAPHY

4.1 Land use Pattern

The land use pattern of the study area is presented in figure 10. Out of the total area, a large part, i.e., about 63.52% (0.108 km²) area is under agricultural land which is now mostly abandoned form due to migration of people from the village. About 18.82% (0.032 km²) area falls under barren land. About 0.018 km² area, which constitutes about 10.20% of the total area is under forest/plants. The settlements in the recharge area cover about 5.29% (0.004 km²) area while about 2.94% (0.005 km²) area is covered by newly constructed road.

4.2 Drinking Water Schemes

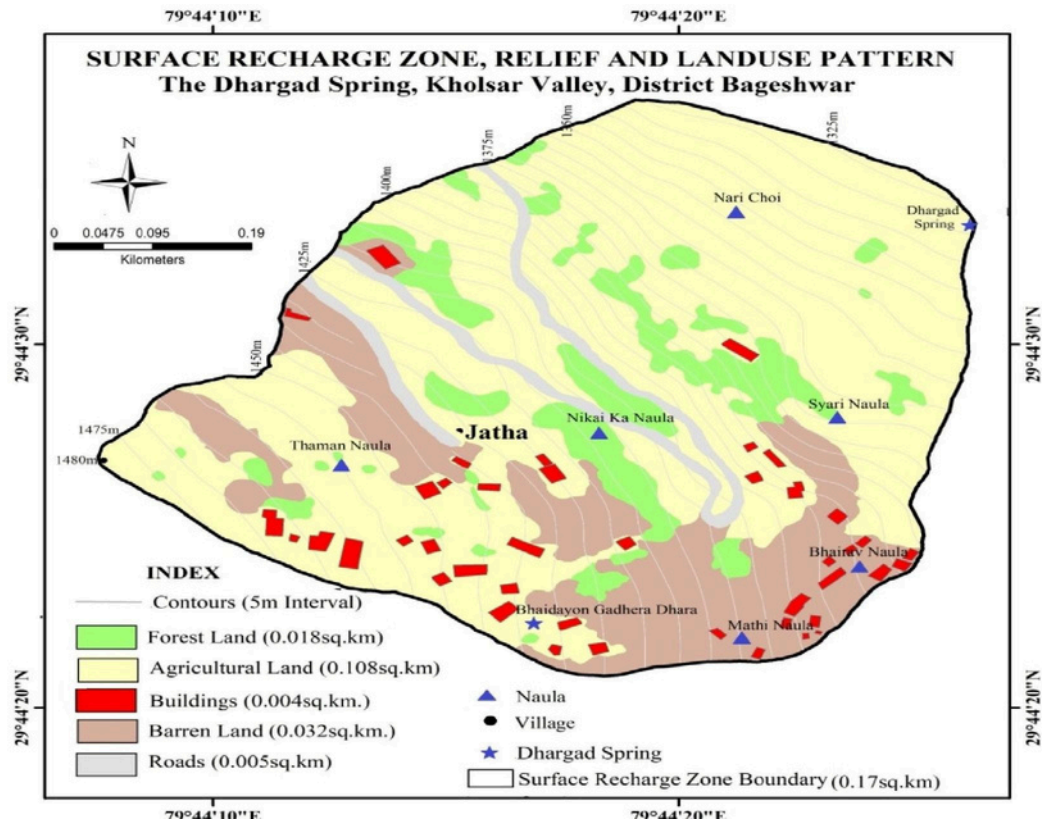


Fig.10: Land use pattern in the experimental recharge area.

5.0 REJUVENATION TREATMENT MEASURES

The treatment measures to be taken for rejuvenation of the spring, naulas and streams (*gadgeras*) of the experimental area are divided into two different types. These are mechanical treatment measures and biological treatment measures. A brief account of objectives of these rejuvenation measures is presented in the following paragraphs.

5.1 Mechanical Treatment

The biological treatment measures take longer time to improve the condition of land for rainwater infiltration, therefore, for speedy groundwater augmentation, mechanical treatment measures are needed in the recharge area. The objectives and types of mechanical treatments to be taken are described in the following paragraphs.

5.1.1 Objectives

5.1.1.1 Particular Objectives

- i) Groundwater augmentation by increasing the rainwater percolation capacity of bed rocks to recharge groundwater storage.
- ii) To increase soil moisture storage in the topsoil, sub-soil and parent material horizons of land.
- iii) To control surface runoff or the problem of flooding.

5.1.1.2 General Objectives

- i) To control accelerated soil erosion.
- ii) To reduce dissolved, suspended, and bed load concentration of streams/rivers water.
- iii) To bring back haryali to nature through water and soil conservation.
-) To improve the quality of economic environment of the local inhabitants.

5.1.2 Treatment Measures

To complete above objectives, the following six structures are recommended for mechanical treatment to fetch the rainwater for groundwater augmentation to recharge springs, naulas and streams. These are: infiltration holes, infiltration trenches, bio-plugs, rill/gully plugs, contour bunds, terrace ponds, check dams and roof rainwater harvesting structures. A brief account of these structures is presented below.

5.1.2.1 Infiltration Holes (Plate-VIII)

These are small holes (size 10-15 cm deep, 5cm Dia and density 5 per/m²) as shown in Plate-VIII, to be constructed on the degraded forest and barren lands to catch and infiltrate the rainwater of the high intensity rainstorms into the soil for groundwater augmentation.



Plate-VIII: An example of infiltration holes (size 10-15 cm deep, 5 cm Dia and density per/m²) constructed to fetch high intensity rain drops in barren as well as in degraded forest to facilitate infiltration of rainwater for groundwater augmentation.

5.1.2.2 Infiltration Trenches (Plate-IX)

These are small cuts (size 1.0 x 0.5 x 0.3 m) as shown in **Plate-IX**, on the hill slopes made at a spacing of 5m to 10m at an average slope condition of 20°. The distance of spacing may be decreased, in case the hill slope is low. These should be constructed in stagger patterns. The purpose of these infiltration trenches is to catch the excess hill slope runoff (overland flow/sheetwash flow) and to facilitate it for groundwater recharge. These infiltration trenches should not be constructed on discharge zones.



Plate-IX: An example of infiltration trench (size 1.0 x 0.5 x 0.3 m) to trap the overland sheet flow from the hill slope for infiltration and to trap the sheetwash erosion sediment.

5.1.2.3 Bio- Plugs (Plate-X)

These are small bio-plugs to be constructed by vegetation and dead wood logs or pine needles and soil across an ephemeral rill/gully. The purpose of these bio-plugs is to break the speed of runoff in the ephemeral rill/gully and to store rainwater to facilitate it for groundwater recharge (Plate-X). The size of these bio-plugs depends upon the local terrain condition.



Plate-X: An example of bio-plug constructed using pine middle across a first order ephemeral stream to break the velocity of runoff, to store water for infiltration in to the aquifer and to check the erosion.

5.1.2.4 Rill/Gully Plugs(Plate-XI)

Due to high intensity rain events, the hill slopes are being eroded steadily through the runoff along ephemeral channels by which deep rills and gullies are being developed. To control this process of riling/gullying, there is need to construct plugs of stones across these rills/gullies to decrease the velocity of stream runoff, to store rainwater for infiltration and to trap the sediment. The size of such plugs (Plate-XI) depends upon the local terrain condition.



Plate-XI: An example of gully plugs across a first order ephemeral stream to check runoff velocity, to store the rainwater and to control process of erosion.

5.1.2.5 Contour Bunds (Plate-XII)

The contour bunds are earthen embankments along contours to capture rainwater in agricultural land (Plate-XII). These help in storing the rainwater for infiltration to groundwater augmentation,

reduce erosion and help in irrigating agricultural crops. These contour bunds should develop in association with the residents.



Plate-XII: An example of series of contour bunds constructed on the edges of agricultural fields to Trap rainwater for irrigation as well as for infiltration to recharge groundwater storage.

5.1.2.6 Terrace Ponds (Plate-XIII)

The terrace ponds are small, artificial water tanks constructed on terraced fields in gentle hillslope areas. These are small shallow tanks with earthen embankments. Their size varies, but typically 10-20 m² in area and 1-2 m deep. The excavated soil of the dig pond should be used for construction of embankments. They are designed to collect and store rainwater through nearby channels or pipes that bring runoff from surrounding catchment areas runoff, providing water for recharge into the aquifer, irrigation and other uses. They also help to control erosion in the hills.



Plate-XIII: An example of terrace pond for harvesting rainwater to groundwater recharge and irrigation of different crops.

5.1.2.7 Check Dams (Plate-XIV)

These are series of small check dams, constructed using cement and iron across second/third order streams having a height of 1.0m to 1.5m (depending upon the local terrain condition of the site). The purpose these check dams (Plate-XIV) constructed across stream at an interval of 30m to 50m

(depending upon slope condition) is to check the flow of intermittent/seasonal streams to facilitate it for groundwater recharge, and to mitigate the process of erosion.



Plate-XIV: An example of series of check dams across an intermittent/seasonal stream to collect water for infiltration and irrigation purposes.

5.1.2.8 Roof Rainwater Harvesting Structure (Plate-XV)

Rainwater harvesting systems may be used to collect and store rainwater from rooftops for domestic use, reducing dependence on external water sources. This system has 4 components:

- i) Roof top collection system: Gutters and downspouts collect and direct rainwater
- ii) Filtration unit: Removes debris and contaminants from the collected water
- iii) Storage tank: Stores the filtered water for later use
- iv) Distribution system: Supplies harvested water for domestic purposes (e.g., flushing toilets, washing clothes, cleaning, irrigation)

This rainwater harvesting system has the following benefits:

- i) Water conservation: Reduces water bills and dependence on external sources
- ii) Reduced stormwater runoff: Decreases burden on drainage systems
- iii) Non-potable use: Suitable for uses that don't require treated water
- iv) Eco-friendly: Promotes sustainable water management practices.

This rainwater system may be used for the following domestic uses such as: i) flushing toilets, ii) washing clothes, iii) cleaning, iv) irrigation, and v) washing utensils/vehicles etc.

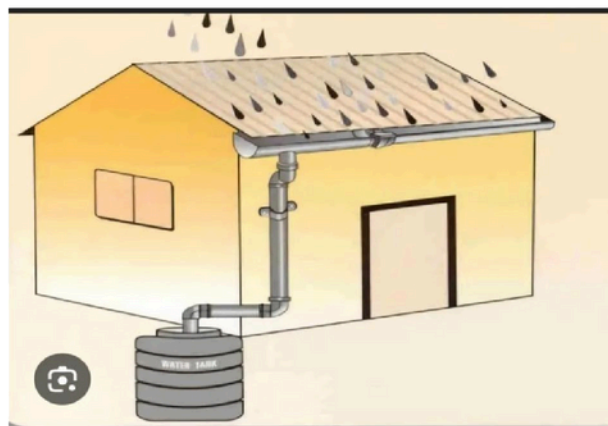


Plate-XV: Rooftop rainwater harvesting structure for kitchen gardening and utensil cleaning etc.

5.1.5.9 Building Rainwater Harvesting for Groundwater Augmentation (Plate-XVI)

Building roof rainwater harvesting structures in hills can help rejuvenate dying springs and streams by augmenting groundwater resources. The system captures rainwater from rooftops, stores it, and allows it to infiltrate into soil and percolate deep into the ground, recharging aquifers. This system has the following 4 components (Plate-XVI).

- i) Roof top collection system: Gutters and downspouts collect and direct rainwater
- ii) Filtration unit: Removes debris and contaminants from the collected water
- iii) Storage tank: Stores the filtered water for later use
- iv) Recharge pit: Allows excess water to percolate into the ground, recharging aquifers.

The benefits of this system are: i) rejuvenates dying springs and streams, ii) augments groundwater resources, iii) reduces soil erosion, iv) conserves water and mitigates water scarcity in hilly areas.

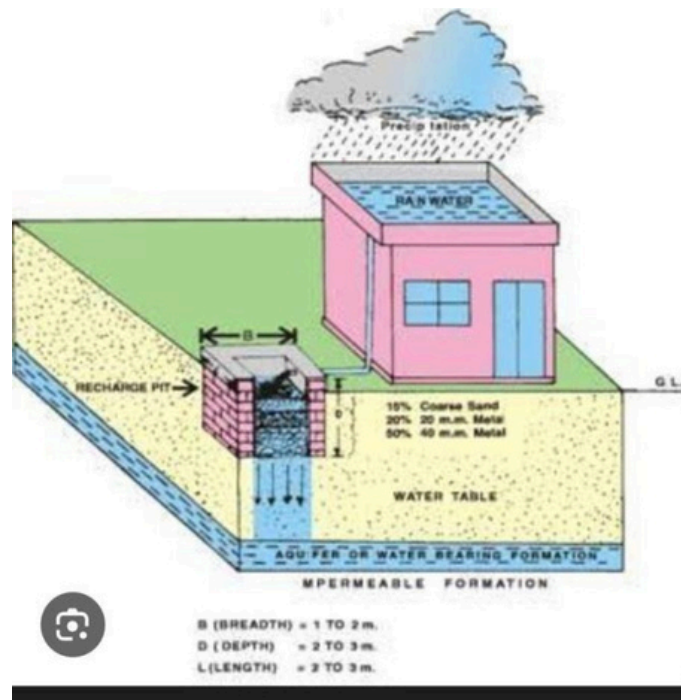


Plate-XVI: Rooftop rainwater harvesting structure for groundwater augmentation for rejuvenation of springs

5.1.5.10 Building Rainwater Harvesting for Groundwater Augmentation (Plate-XVII) Roof rainwater harvesting structures in hills can serve multiple purposes also, including domestic use, groundwater augmentation, and rejuvenation of dying springs and rivulets. The system captures rainwater from rooftops, stores it, and utilizes it for various needs. This system has five components (Plate-XVII):

- i) Roof top collection system: Gutters and downspouts collect and direct rainwater
- ii) Filtration unit: Removes debris and contaminants from the collected water
- iii) Storage tank: Stores the filtered water for later use
- iv) Recharge pit: Allows excess water to percolate into the ground, recharging aquifers.
- v) Distribution system: Supplies harvested water for domestic purposes (e.g., irrigation, toilet flushing, cleaning).

There are different benefits of this system as mentioned below.

- I) Domestic use: Provides water for non-potable purposes, reducing dependence on external sources
- i) Groundwater augmentation: Recharges aquifers, rejuvenating dying springs and rivulets.
- ii) Soil conservation: Reduces soil erosion and landslides
- iii) Water scarcity mitigation: Helps alleviate water scarcity in hilly areas
- iv) Eco-friendly: Promotes sustainable water management practices,
- v) Rejuvenation of Dying Springs and Rivulets: Recharged groundwater feeds into springs and rivulets, restoring their flow, and
- vi) Helps maintain aquatic ecosystems and supports biodiversity

Thus, by implementing rainwater harvesting structures in hills, communities can ensure a sustainable water supply, conserve natural resources, and promote eco-friendly practices.

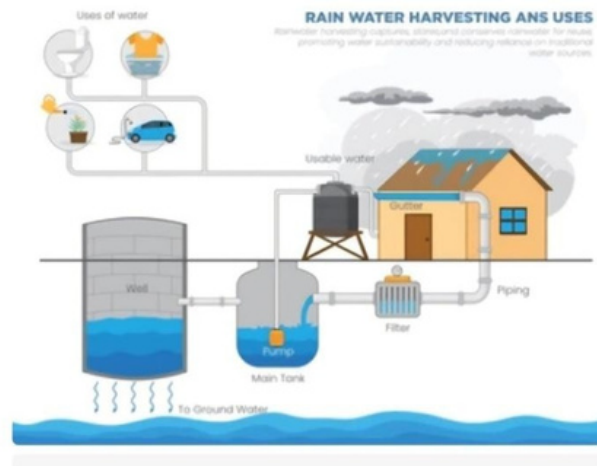


Plate-XVII: Rooftop rainwater harvesting structure for groundwater augmentation for rejuvenation of springs and for different domestic purposes

5.2 Biological Treatment

Biological treatment measures include –afforestation, plantation of trees, shrubs and grasses, development of nurseries, horticulture and floriculture. The past experiences reveal that, no biological treatment work should be done for rejuvenation of dead/dying water resources without ensuring complete protection from fire as well as from damages through animals and human beings. A brief account of objectives and treatment measures of biological treatment is presented below.

5.2.1 Objectives 5.2.1.1 Particular Objectives

- i) To increase the rainwater infiltration capacity of land for soil moisture storage.
- ii) To increase groundwater storage by deep percolation of rainwater into rocks on long term basis.
- iii) To decrease surface runoff.

5.2.1.2 General Objectives

- i) To control accelerated erosion.
- ii) To mitigate the problem of flooding on long term basis

- iii) To control the process of masswasting, i.e., landcreeping, slumping, landsliding etc.,
- iv) To regenerate the degraded natural environment.
- v) To improve the quality of economic environment of the local inhabitants.

5.2.2 Treatment Measures

To meet above objectives, biological treatment measures such as afforestation, plantation, and development of horticulture and nurseries should be done. A brief account of all these biological treatment measures is presented in the proceeding paragraphs.

5.2.2.1 Afforestation Afforestation means establishing a forest in an area where there was no previous tree cover. This involves planting of native species, allowing natural regeneration, and promoting biodiversity. This may be done on degraded land, like barren areas or abandoned agricultural lands. It is suggested that in such available areas as the afforestation of water harvesting (broad leaves) species, local shrubs and grasses such as Napier and vertebrales should be done by initiating massive social forestry programmed in the recharge areas with close association with local community and the Forest Department and Watershed Management Department.

5.2.2.2 Plantation This is a process of establishing a forest plantation in an area often with a single species (monoculture). Typically, plantation involves non-native or commercial species for timber, pulp, or other products. This work may be done in association of the Forest Department. **5.2.2.3**

Horticulture It includes to development of community horticulture and individual horticulture programmers specially on the barren and abandoned agricultural lands in the recharge areas by which the waste land of the watershed could be carpeted by plants to increase rainwater infiltration capacity of land. This work should be done in association with the local inhabitants and the Department of Horticulture and Watershed Management. This will also improve the economic base of the local inhabitants.

5.2.2.4 Development of Nurseries It includes development of nurseries of broad-leaved plants, fruits, herbs and grasses etc. for conducting social forestry, community/individual horticultural and herbal gardening programmers in the recharge zones. This work should be done in association of the local community and the Department of Horticulture and forest.

5.2.2.5 Development of Horticulture/Floriculture For economic development of the area, development and horticulture and floriculture should also be started with the guidance of the department of horticulture and floriculture. Beside economic benefit these activities shall accelerate the process of rainwater infiltration and deaccelerate the process of erosion.

6.0 MICRO-PLANS FOR REJUVENATION INTERVENTIONS

Micro-plan maps (Fig.12,13,14 and 15) of the experimental area depict the notional locations/sites of mechanical and biological treatment measures. These maps may work as a faithful guide for implementation of micro-plans in field. A brief micro-plan related to proposed mechanical and biological treatments is presented in the following paragraphs.

6.1 Mechanical Treatment

Based on the standards of different mechanical treatment measures, geomorphological and geographical characteristics, numbers of each mechanical treatment measure were worked out which are presented in Table-3. A brief account of each mechanical treatment measure is described in the following paragraphs.

Table-3: Estimated numbers of mechanical treatment measures to be constructed in the experimental recharge area

Name of Mechanical Measures	Number of Treatment Mechanical Measures			
	Forest Land	Barren Land	Abandoned Agricultural Land	Agricultural Land
Infiltration Holes	90000	160000	150000	-
Infiltration Trenches	291	2050	-	-
Bio-plugs	-	-	20 across the upper reaches of two rills/gullies (see location of these rill/gullies in Fig.13)	
Rill/Gully –Plugs	-	-	5 across in the lower reaches of two rills/gullies (see location of these rill/gullies in Fig. 13)	
Terrace Ponds	-	2	10	20
Contour Bunds	-	-		5000 running meters
Check Dams	-	-	2 near the mouth of two rill/guiles (see location of these rill/gullies in Fig.12).	

6.1.1 Infiltration Holes

Infiltration holes are small holes (size 10-15 cm deep, 5cm Dia and density 5 per/m²) as shown in **Plate-VIII**. The purpose of these infiltration holes is to catch the excess hill slope runoff caused by high intensity rain drops and to facilitate it for groundwater recharge. These infiltration holes should be constructed in barren land, forest land and in abandoned agricultural land. Figure 11 depicts the notional locations of these proposed infiltration holes which must be constructed as much as 90,000 in forest land, 160,000 in barren land, and 1,50,000 in abandoned agricultural land (Table-3). These holes may be filled after 2 monsoon rains, then new holes must be constructed. This work should be done in association with department of Forest and Watershed Management.

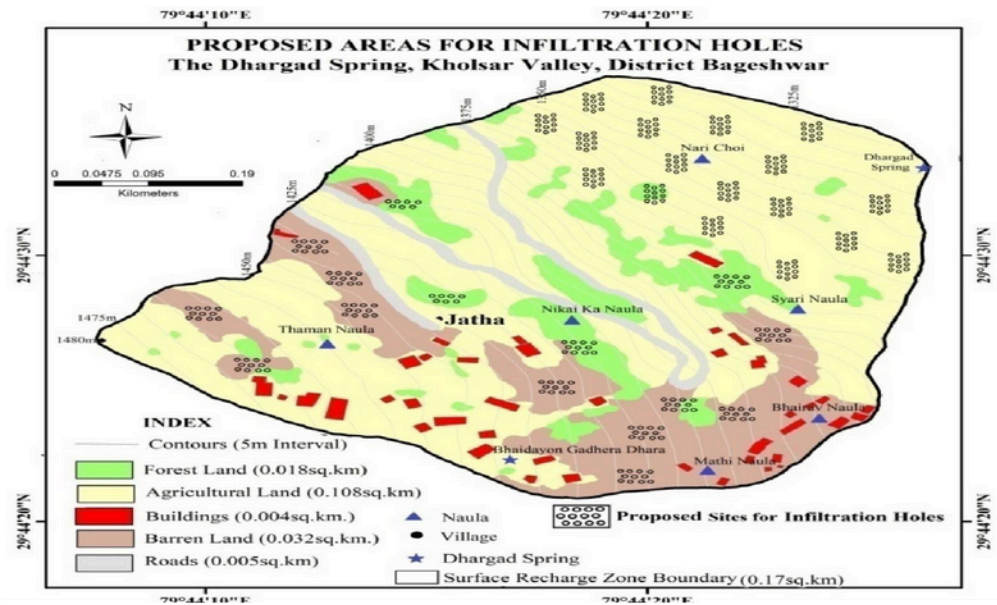


Fig.11: Location of proposed sites for Infiltration Holes.

6.1.2 Infiltration Trenches

Infiltration trenches are small cuts (size 1.0 x 0.5 x 0.3 m) as shown in Plate-IX, on the hill slopes made at a spacing of 5m to 10m at an average slope condition of 20°. The distance of spacing may be decreased if the hill slope is low. These should be constructed in stagger patterns. The purpose of these infiltration trenches is to catch the excess hill slope runoff caused by high intensity rain drops and to facilitate it for groundwater recharge. These trenches must be constructed in the degraded forestland and barren lands preferably on convex hillslopes of barren land, degraded forest (Fig.12). A total of about 2341 trenches must be constructed, i.e., 291 in forest land and 2050 in barren land. These trenches may be filled after 3-4 monsoon rains. After filling it completely, a plant should be planted in this filled trench and new trench should be constructed in place of the filled trenches. This work should be done in association with department of Forest and Watershed Management.

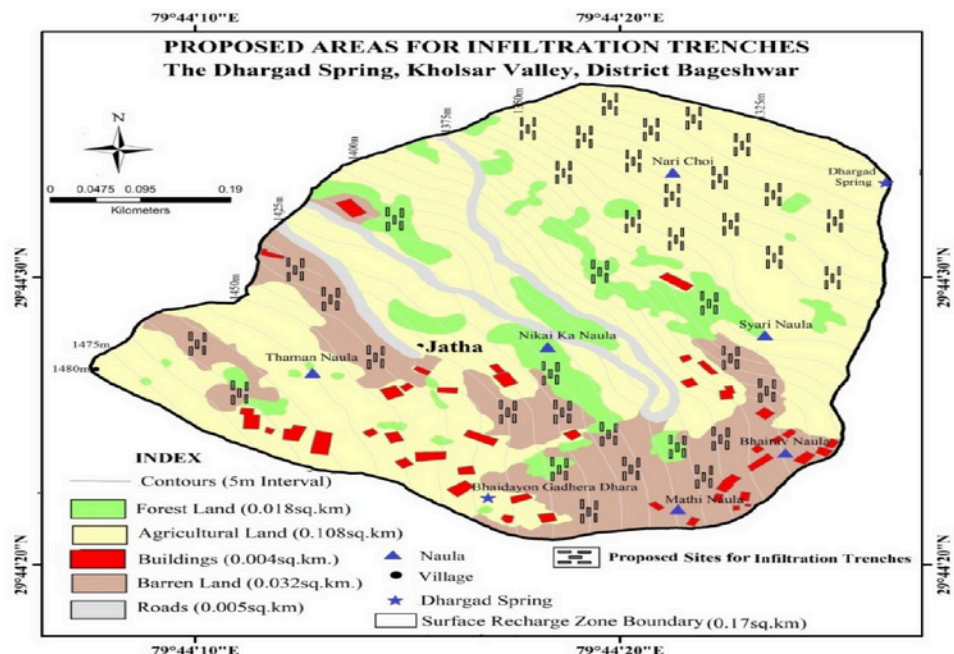


Fig.12: Location of proposed sites for Infiltration Trenches.

6.1.3 Bio- Plugs

These are small bio-plugs to be constructed by vegetation leaves and dead wood logs or pine needles and soil across an ephemeral rill/gully. The purpose of these bio-plugs is to break the speed of runoff in the ephemeral rill/gully and to store rainwater to facilitate it for groundwater recharge (Plate-X). The size of these bio-plugs depends upon the local terrain condition. In the study area, there are two rills/gullies. About 20 bio-plugs at a vertical interval shall be needed across in the upper reaches of both the rills/gullies (Fig.13). This work should be done in association with department of Forest and Watershed Management and Soil Conservation.

6.1.4 Rill Plugs/Gully Plugs

Due to high intensity rain events, the hill slopes are being eroded steadily through the runoff along ephemeral channels by which deep rills and gullies are being developed. To control this process of rilling/gullying, there is need to construct plugs of stones across these rills/gullies to decrease the velocity of streams, to store rainwater for infiltration and to trap the sediment. The size of such plugs (Plate-XI) depends upon the local terrain condition. In the study area 5 gully plugs shall be needed in the middle reaches across 3 ephemeral channels (Fig.13). This work should be done in association with department of Watershed Management and Soil Conservation.

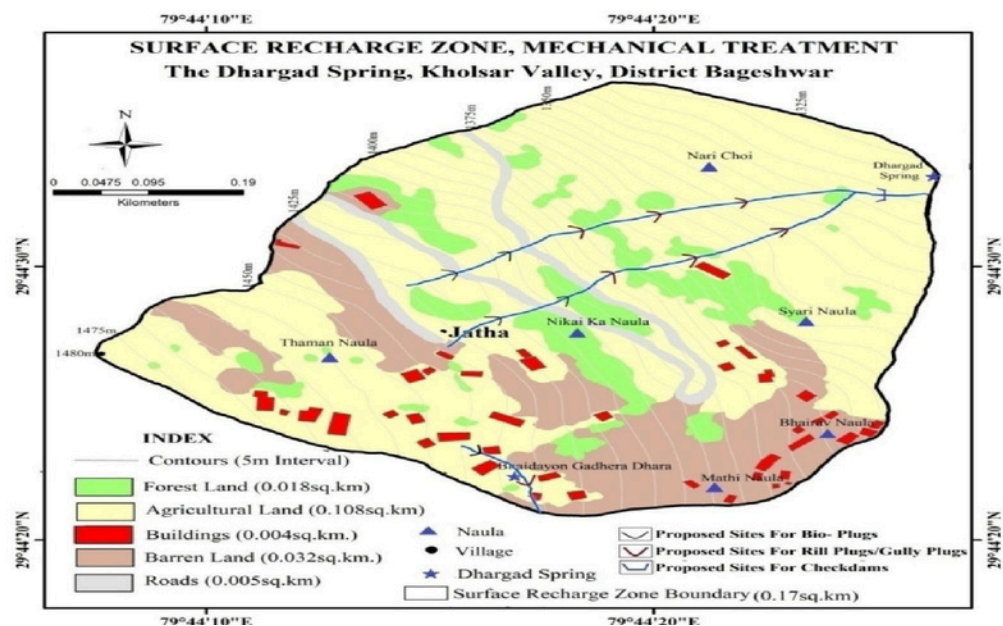


Fig.13: Location of proposed sites for Bio-Plugs, Rill-Plugs and Check dams.

6.1.5 Contour Bunds

The contour bunds are earthen embankments along contours to capture rainwater in agricultural land (Plate-XII). These help in storing the rainwater for infiltration to groundwater augmentation, reduce erosion and help in irrigating agricultural crops. These contour bunds should develop at the edge of agricultural land in association with the residents. As per rough estimate about 5000 running meters contour bunds shall be needed in the study area. Figure 14 depicts the notional locations of these contour bunds.

6.1.6 Terrace Ponds

Terrace ponds are small, artificial water tanks constructed on terraced fields in gentle hillslope areas. These are small shallow tanks with earthen embankments (Plate-XIII). Their size varies, but typically 10-20 m² in area and 1-2 m deep. The excavated soil of the dig pond should be used for construction of embankments. They are designed to collect and store rainwater through nearby

channels or pipes that bring runoff from surrounding catchment areas runoff, providing water for recharge into the aquifer, irrigation and other uses. They also help to control erosion in the hills. In the barren land 10 and in the agricultural 20 terrace ponds may be constructed in the study area. Figure 14 depicts suitable sites for construction of terrace ponds in the study area which may be constructed in association with the local inhabitants.

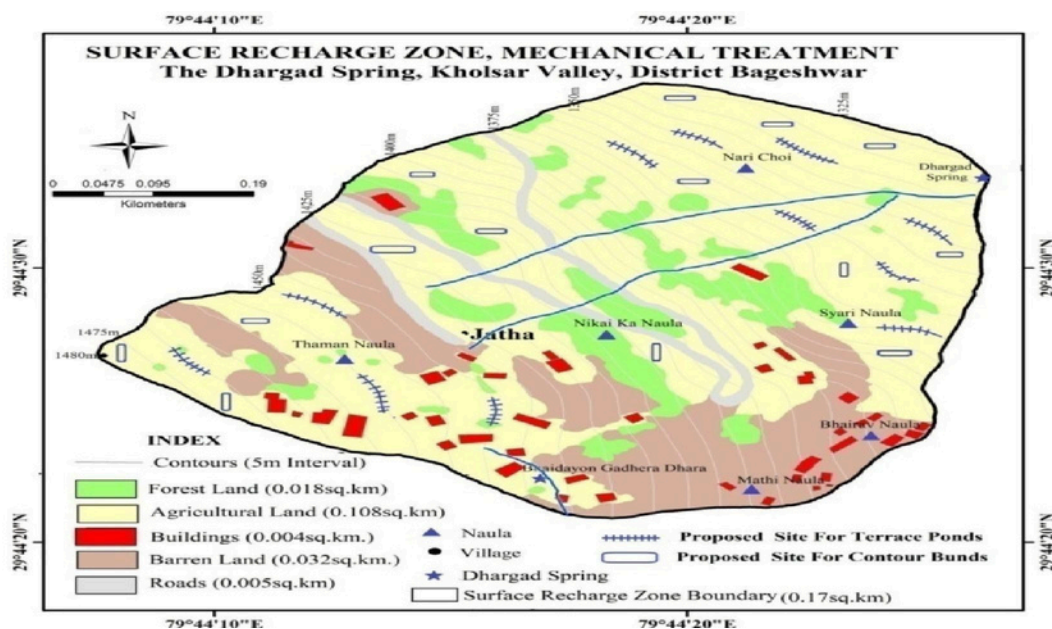


Fig.14: Location of proposed sites for *Contour Bunds and Terrace Ponds*.

6.1.7 Check Dams

These are series of small check dams, constructed using cement and iron across second/third order streams having a height of 1.0m to 1.5m (depending upon the local terrain condition of the site) (Plate- XIV). The purpose of these check dams is to check the flow of intermittent/seasonal streams and to store

the rain water to facilitate it for groundwater recharge, irrigation and to mitigate the process of erosion. In the study area two such check dams may be constructed at the mouth of two rills/gullies (Fig. 13), if needed absolutely.

The roof rain water harvesting structures are of two types. The first one is roof rainwater harvesting for domestic use only (Plate- XV) and roof rainwater harvesting for ground water augmentation and for both groundwater augmentation and domestic purpose (Plate-XVI and XVI).

6.2 Biological Treatment

6.2.1 Afforestation and Plantation

Afforestation means establishing a forest in area where there was no previous tree cover. This involves planting of native species, allowing natural regeneration, and promoting biodiversity. This may be done on degraded land, like barren areas or abandoned agricultural lands. It is suggested that in the study area:

- i) about 0.032 km² land which constitutes about 18.12% of the study area is under barren condition. The spatial distribution of this barren land is presented in figure 6. This entire barren land may be used for i) afforestation of water harvesting (broad leaved) species, local shrubs and grasses such as Napier and vertebrae; and ii) plantation of a single

- species (monoculture) and non-native or commercial species for timber, pulp, or other products. This work may be done in association of the Forest Department. About 0.032km² of the study area which most of the part is as abandoned barren condition (Fig.11) which may be used for individual and community based horticultural farming in association with local community. This work should be done in association with the Department of Horticulture. About 0.132km² land is under agricultural and barren condition in which on suitable sites nurseries of: i) broad leave plants, local shrubs and grasses; ii) different herbal and ornamental plants; iii) different fruits; and flowers for floricultural development can be done in association with department of Forest and Horticulture. should be done by initiating massive social forestry programmer in the recharge areas with close association with local community and the Forest Department and Watershed Management Department.
- ii)
 - iii)
 - iv)

6.2.2 Development of Horticulture/Development of Floriculture

he experimental recharge area is well connected by road as the road passes from the centre of the village, therefore there is good potentiality of development of horticulture as well as floriculture development in the experimental recharge area. The village panchyat should develop individual; and community horticultural plans and should identify land for implementing these plans, and should contact to the department for implementation. Similarly, the floriculture development plan should be done in association and guidance of the department of floriculture.

ACKNOWLEDGEMENT

In a workshop organized by the AMAN NGO, Almora in the month of November, 2025 at the Kasar Devi (Almora), one of the authors of this report (JSR) got an opportunity to make a detailed presentation on “River Rejuvenation Strategy of the Rain fed Rivers of the Uttarakhand State”- among a group of Experts of different NGOs of the Uttarakhand State. Thanks to the AMAN NGO for providing me this opportunity, because this report is the outcome of the motivation which I got after deliberations and discussions with the grassroots level stakeholders. Thanks are due to the Centre of Excellence for Natural Resources Data Management System in Uttarakhand for technical support and the AMAN NGO for all logistic support to prepare this report on a representative rainfed watershed of Uttarakhand, in particular and Himalaya, in general..

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ANNEXURE -I

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fd;kx;kgSATuleqnk; dk sifj; ktsuk dh rduhd ,o adk;kZUo;ua dk sle>ku sd sfy, fnukda
29&11&2025dksxzke&tkBk ,oa djklhcqxak e saty lja{k.k leqnk; vfHkeq[khdj.k izf''k{k.k fd;k
x;kA

mnn~''s ;

- ☐ tulgHkkfxrk ls xzke tkBk ,oa djklhcqxak ds ty lk=sk sadk siquthZfor djuka
- ☐ ty lk=sksa ds iquthZou d sfy, ty foKkuh MkW0 thou flga jkor dh fjiksV Zdks leqnk; d s
lkFk lk>k djuka
- ☐ tuleqnk; dks ty lks= iquTuZu ds fy, oSKkfud rduhd l svoxr djuk rFkk tkx:d
djuka

iFzke izf''k{k.k&

Jhtxnh''kifj; ktsukQhYMleUo;d}kjkizfrHkkfx;ksarFkklUnHkZO;fDrdkLokxr
fd;kx;kAveudhdk;dZkjhleUo;duhfyeKHKV~V}kjkdk;dZzedslUnHkZ,oamn~''s'';ksaij
ppkZ dh xbZA ppkZ ds nkSjku lenq k; ls fuEu leL;k,a fudy dj vkbZ%&
xfeZ;ksaesa2&3eghusikuhdhxHakhjleL;kjgrhgSA
tyok;qesaifjorZugksjkgSaA
?kjksaesaugSa]yfsduxfe;Zksaesaalks=l[wkusyxrgSrFkkikuhdkladVcutkrkgSA

rRi''pkr~lUnHkZO;fDrMkW0thouflagjkorth}kjktyldaVvkSjmlsdskj.kksaij
foLrkjlsle>k;kx;kAMkW0jkor}kjk;gcrk;kx;kfdoreZkuesatyok;qifjorZrrFkk
ughadjikjghgS]ftldkj.ktylda yU;dkj.kksalsvf;fero''kkZgksjghgSrFkktxayksavkSj[ksrksadsek;/els/kjrhtyvo''kkffs'kr
VdhleL;kmRiUugksjghgSAIkFkghmUgksuasfoLrkj
lsckrsgq,fdiNysdqNo'kksZaesaigkMk+saesaihiy]mrhl]ckWltlSspkSMhifRrokysò`{kksadh
Ifa;kdegksjghgS,oalr>M+¼lkrfnurdyxkrkjgksusokyhfjef>eckfj''k½dsLFkkuij
elwyk/kkjo'kkZgksusdsdkj.kHkhgekjstyk=s detksjgksjgsgSAa tc?kusiMs+ughacprsrFkk
ckfj''kdkikuhl/kscgtrkgSvkSj/kjrhesatyvo''kkfs'krughagksikrkrksxfeZ;ksaesylyks=
tYnhl[wktrkgSAmUgksuasxkeh.kksadksle>krsgq,dgkfd&^^gesafprakvktfdfy,ugh]a
cfYd 20 lky ds ckn dh gSâ ^^A

ifj; kts tSlk fd vki tkur sgS veu lLaFkk d slkFk feydy vki ykxsks a}kjk tyok;q laj{k.k
ukdk;dZze rS;kj fd;k x;k gSA bl ifj;ktsuk esa vkids xzke ds tylk=sksa ds iquthZou ds
fy,vxfzyf[kr rduhd iz;kxs es aykbZ tk,xah%&

1- Infiltration Holes ¼ty fjlko grsq /kjr ij fd, tku soky sNns½

10&15 lesh xgj]s 5 lseh Mk;ehVj oky sNksVs xM~Ms] ftudh njwh yxHkx 5 izfr

oxZehVj gksxh A

igkMh <yuku aij NksVs&NkVss xM~Ms [kksndj muea so'kkZ dk ikuh jkds tehu esa

Hkstk tk ldrk gSA

bu xM~Mks adks lkekU;r% lEHky l scuk;k tkrk gSA

; sxM~Ms feV~Vh dh ueh dk sc<kr sgSa vkSj Hkwfexr ty dks c<krs gSAa

2- Infiltration Trenches

□ bUfQYV^as''ku V^asap (1.0 × 0.5 × 0.3 m) rFkk 10 ehVj dh njwh e scuk, tk,xasA

□ igkMh <yku dh fn''kk e saContour trenches cukdj o'kkZ dk cgko /khek gksrk gSA

□ ikuh :ddj tehu esa lek tkrk gS] ftll suhp sds lHkh lks= etcwj gksr sgSaA

Bio-Plugs

□ ;g ikfjLFkfrdh rU= dks cuk, j[krk gSAa

□ bUg s4 bpa xgikbZ es acuk;k tkrk gS rkfd tc cjlkr dk ikuh rhoz cgko ls vk; s
rk sikuh sjkdsk tk ldAs

□ bUg safi:y ;k ?kkl ls cuk;k tk ldrk gS] fi:y ,oa ?kkl dk sjLlh l scka/kk
tkrk gSA

□ ;g cjlkrh ukyksa dh rhozrk dk sde djr sgS rFkk feV~h ds dVko jkdsrk gS rFkk
ty dk stehu ds vUnj y stkrk gSA

3- Contour Bunds

[krs ds uhps dh rjQ dseMksadks Åpak fd;k tkrk gS rkfd cjlkr dk ikuh

[krksaesa:ddj /kjr esavo''kksf'kr gksldAs

dVawj cMa~l [ksrksa ;k <yukuaij leksPp js[kkvksads vulqkj cuk, tkusokysyEcs
feV~h ds ca/k gSAa

ifj;ktsuk dh rduhdksadksle>kkusdsmijkUr MkW0 jkor }kjk tu leqnk; dks
crk;k fd ty lks=ksadks lqjf[kr j[kus ds fy, dsoy xM~M]s Vs^aap vkSj cjlkrh ukyksa dksidMuk
gh i;kZlr ugha gSA bl dsckn lclsegRoi.wkZ ctaj Hkwfe esa o`{kkjkis.k djuk] tks bl izdkj gSA

igkMks aij LFkkuh; iztkfr;ksatSl&mrhl] ckWt] ihiy ,oa cqjkWl vkfn dsiMs+ ty laj{k.k.k
dsfy, lcls izHkkoh gSA

bu iMksadkh tMs+ ckfj''k dk ikuh jksdsdj mls/khj&s/khjs tehu esa Hkstrh gSAa

blsu dsoy Spring etcrw gkrss gS] cfYd feV~h dk dVko Hkh :drk gSA

Floriculture ¼Qywsadh [krsh½ dk lq>ko Hkh fn;k x;k] ftlds fy, xzkeh.kksadksizLrko
r;Skj djusdsfy, dgk x;kA

lq>kko %&

cSBd d snkSjku uhfyek HkVV~ }kjk Li'V fd;k x;k fd veu lLaFkk orZeku es avkid s xkWo e saty laj{k.k d sfy, dk;Z dj jgh gSA o`{kkjkis.k ,oa Qwyksa dh [ksrh tSl sdk;ksZa dks lQy cuku sds fy, xzke leqnk; dk Lo&fodflr iz;kl vko";d gSA ¶lykshdYpj ,oa o`{kkjkis.k dk s dk;Z ljdkj ds lg;ksx ls fd;k tk,xk bl izfdz;k esa veu laLFkk rduhdh elxZn"KZu] ;ktsuk cuku sesa vko";d lg;ksx iznku djsxahA veu lLaFkk }kjk bu dk;kZs ads fy, foRrh; lg;ksx djuk IEHko ugh agSA vki ykxs Lo;a o`{kkjksi.k ,o aQywks adh [ksrh d sfy, izLrko rSS;kj djsxasAa

xzke esa ifj;ktsuk dh rduhdks adk sle>kus ,oa xkzeokfl;ksa dh leL;k,a lquu sd s mijkUr cSBd dh foLr`r ppkZ vkSj xzkeh.kksa dh lgefr] l>qkko ds mijkUr lkefwgd :i ls ;g egRoiw.kZ fu.kZ; fy;k x;k fd xzke tkBk d sokMZ uEcj&6 ,oa okMZ uEcj&7 esa ty laj{k.k l s lEcU/kr dk;Zdks izkFkfedrk ds vk/kkj ij "kq: fd;k tk,xkA

f}rh; izf"K{k.k

ifj;kts f}rh; izf"K{k.k xzke&djklhcxqak e safd;k x;k] ftldk izkjEHk Jh txnh"k }kjk xzkeh.kks a ,o ae[q; izf"K{k.kd d sLokxr d slkFk fd;k x;kA MkW0 thou flga jkor }kjk tyok;q lja{k.k uk ds mn~n"s;ksa ,oa dk;kZsa dks tu leqnk; dks foLrkjiwoZd crk;k x;kA lFk gh muds }kjk dkslh unh esa vius }kjk fd; sx; sdk;kZsa dk mnkgj.k Hkh fn;k vkSj veu dh dk;Zdkjh leUo;d uhfyek HkVV~ }kjk dk;dZe d sfo'k; ,oa mnn~"s;ksa ij ppkZ dh xbZ rnkisjUr MkW0 thou flag jkor }kjk tyok;q ijh{k.k ifj;ktsuk dh rduhdks dks tu leqnk; dks le>k;k x;k] tk sfuEuor~ gS%&

1- Infiltration Holes

xM~Ms dk vkdkj%& 10&15 lesh xgj]s yxHkx 5 izfr oxeZhVjA ; sNksV&sNkVss xM~M sgkrss gS] ftuesa ikuh fjldj uhp stkrk gSA blI stehu dh ty&/kkj.k {kerk c<rh gS vkSj Hktwy fjpktZ gksrk gSA xM~Mks adk scuku sdk mn~ns"; gS fd ckfj"K dk ikuh tehu e saT;knk l sT;knk vo"Kksf'kr gks ldsA

2- Infiltration Tranches

- vkdkj&(1.0 × 0.5 × 0.3 m) |
- bu Vªspals dk svkerkSj ij 10 ehVj dh njwh ij cuk;k tkrk gS] rkfd cM s{k=s e sa ikuh fjpktZ gk sldsA
- ; syEc svkSj ldaj s[kksn sx, uky sgksr]s ftUgsa ckfj"K ds ikuh dks tehu esa vo"Kksf'kr djus ds fy, cuk;k tkrk gSA
- budk eq[; mnn~s"; gS feVV~h dh ty&lxyz {kerk c<k+uk vkSj ikuh dk scgus l s cpkuk gSA

3- Check Dam

- pds MSe ,d NkVsh ckW/k tSlh jpuk gksrh gS] tk sfdlh NksV suky]s x/kjss ;k cijlkrh ikuh ds jkLr sesa cukbZ tkrh gS] rkfd%&
- ikuh dk cgko /khek gks tk, rFkk ikuh tehu e sa/khj&s/khj slek ldsA

- vkl&ikl d sLkk=s ,oa /kkjk sae saikuh dh vkiwfr ZdH tk ldAs
- feW~h ds dVko dks de fd;k tk ldAs

izf''k{k.k ds lekiu ls iwoZ rFkk MkW0 jkor d sifj;ktsuk dh rduhdks adks le>kku s
d si''pkr uhfyek HKV~V }kjk xkzeokfl;k sal slq>ko fy;k x;k fd ge vkid s{k=s e saifj;kstuk d s
dk;kZsa dks dc l sizkJEHk dj ldr sgS] fti ij xkzeokfl;k sau sloZlgefr l scrk;k fd ge vius
xkze esa ifj;ktsuk d sdk;Z dks "kh?kzkfr"kh?kz izkJEHk djxsaAs
lyaxud

- 1 izfrHkkxh lpwh 1-1&tkBk j1-2& djlhcWqxkA
- cSBd QkVskS

2

नाम	गैव	हस्ताक्षर
विषय - जल स्रोत रिचार्ज के लिए समुदाय बैठक		
आज दिनांक 29/11/2025 को ग्राम सभा जालम में जल स्रोत रिचार्ज के लिए एक बैठक की आयोजित की गई जिसमें ग्राम जालम के 28 प्रतिभागियों की भागीदारी रही। बैठक में अमन वांस्वा की तरफ से कार्यकारी अधिकारी व जगदीश जी उपस्थित थे।		
(1) आनन्द पलाय	जालम	आनन्द पलाय
(2) गीता	"	गीता
(3) जलुली देवी		
(4) देवकी देवी		
(5) पप्पू लाल		पप्पू लाल
(6) गणेश राम		गणेश राम
(7) जगदीश पलाय		जगदीश पलाय
(8) अर्जुन लाल		अर्जुन लाल
(9) सुरज		सुरज
(10) हरीश राम		हरीश राम
(11) गोपाल राम		गोपाल राम
(12) अमरा देवी		अमरा देवी
(13) आनन्दी देवी		

(14) बसंती देवी	
(15) विमला देवी	विमला देवी
(16) गोपाल राम	गोपाल राम
(17) गंगा देवी	गंगा देवी
(18) गीता	गीता
(19) अमरा देवी	अमरा
(20) लुलसी देवी	
(21) गंगा देवी	
(22) प्रेमा देवी	प्रेमा देवी
(23) जगदीश पलाय	जगदीश पलाय
(24) भैरव राम	भैरव राम
(25) श्रीमानन्द	श्रीमानन्द
(26) गणेश राम	गणेश राम
(27) लुलसी देवी	लुलसी देवी
(28) दिनेश चन्द	दिनेश
(29) अर्जुन सिंह	अमन
(30) जगदीश	जगदीश

- विषय - आज दिनांक 29/11/2025 को काफली गैर (कराखीला) में जल स्रोत रिपार्च के लिए समुदाय के साथ एक बैठक आयोजित की गयी। बैठक में उच्च प्रतिभागियों की उपस्थिति रही अमन खेड़िया की ओर से काफली समुदाय की अध्यक्षता की गई थी।

सं	नाम	हस्ताक्षर
(1)	हरीश नगरकोटी	महेश
(2)	नवीन	विनय सिंह
(3)	सुन्दर सिंह	सुन्दर सिंह
(4)	लाला सिंह नगरकोटी	लाला देवी
(5)	कमला देवी	दीपा देवी
(6)	श्रीवा (दीपा) देवी	गोमा देवी
(7)	गोपाल सिंह	गोपाल सिंह
(8)	जिवन देवी	जिवन देवी
(9)	गोपाल सिंह नगरकोटी	गोपाल सिंह
(10)	श्रीवा देवी	श्रीवा देवी
(11)	गंगा देवी	गंगा देवी
(12)	दीपा देवी	दीपा देवी
(13)	सुम्मा देवी	सुम्मा देवी
(14)	हमा देवी	हमा देवी
(15)	हमा देवी	हमा देवी
(16)	जानकी देवी	जानकी देवी
(17)	रेखा देवी	रेखा देवी
(18)	वीना देवी	वीना देवी
(19)	श्रीवा देवी	श्रीवा देवी
(20)	रेखा देवी	रेखा देवी
(21)	उमेशा देवी नगरकोटी	उमेशा देवी
(22)	वीना नगरकोटी	वीना नगरकोटी
(23)	अश्वती नगरकोटी	अश्वती
(24)	गंगा देवी	गंगा देवी

(25)	ललिता देवी	ललिता देवी
(26)	हमा देवी	हमा देवी
(27)	विमला देवी	विमला देवी
(28)	मंग देवी	मंग देवी
(29)	पारुषी देवी	पारुषी देवी
(30)	ललिता देवी	ललिता देवी
(31)	शान्ति देवी	शान्ति देवी
(32)	रमा देवी	रमा देवी
(33)	विनीता देवी	विनीता देवी
(34)	अमन सिंह	अमन
(35)	जगदीश	अमन





